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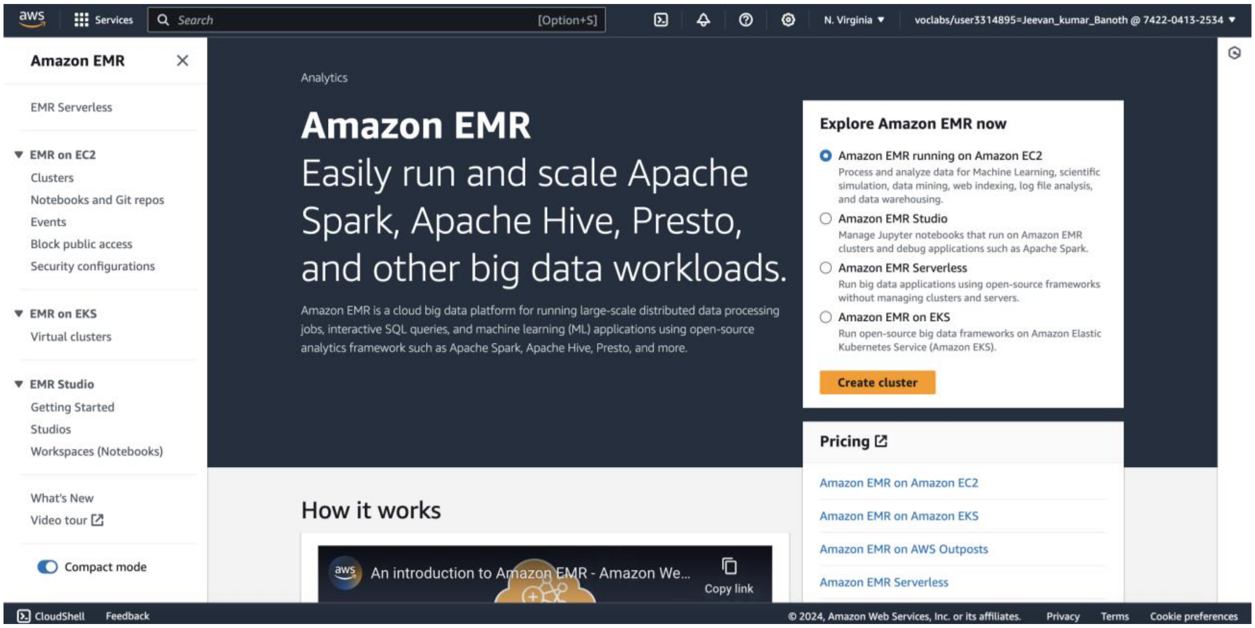
CIS 602-7103: Special Topics in CIS

Homework – 5:

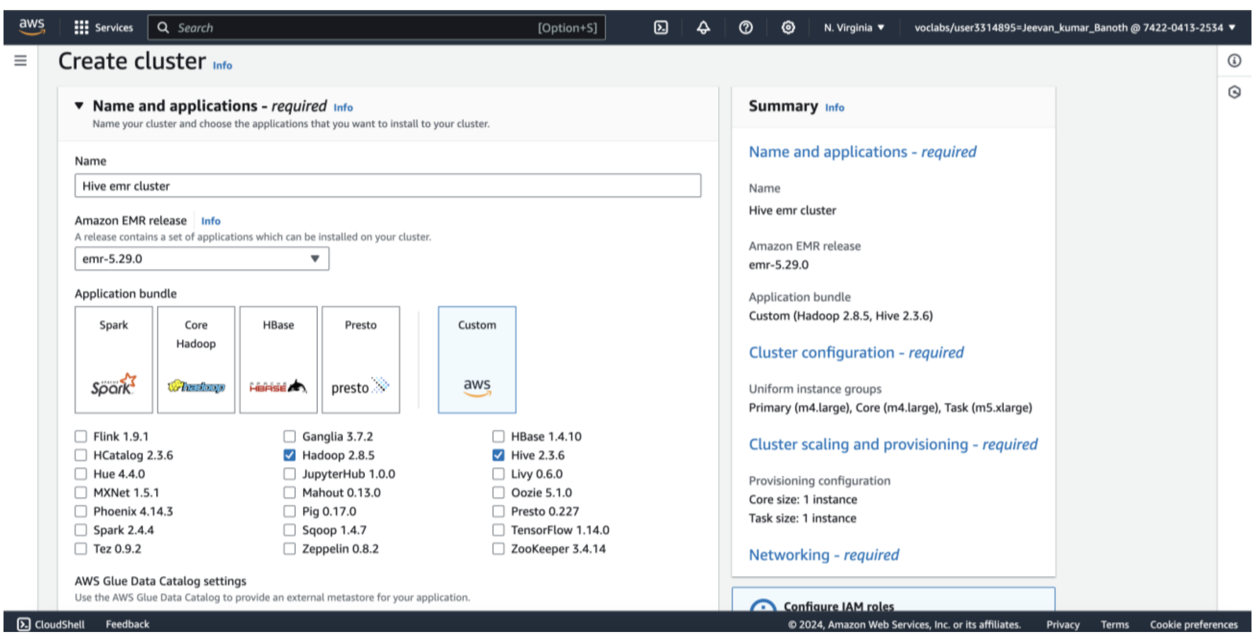
Task 1: Install an Amazon EMR Cluster.

To find EMR, use the Services search box in the AWS Management Console. Look out for the EMR entry on the list.



Choose "Create cluster" from the menu.

Click the Go to advanced choices button located in the upper right corner of the Create Cluster - Quick choices screen.



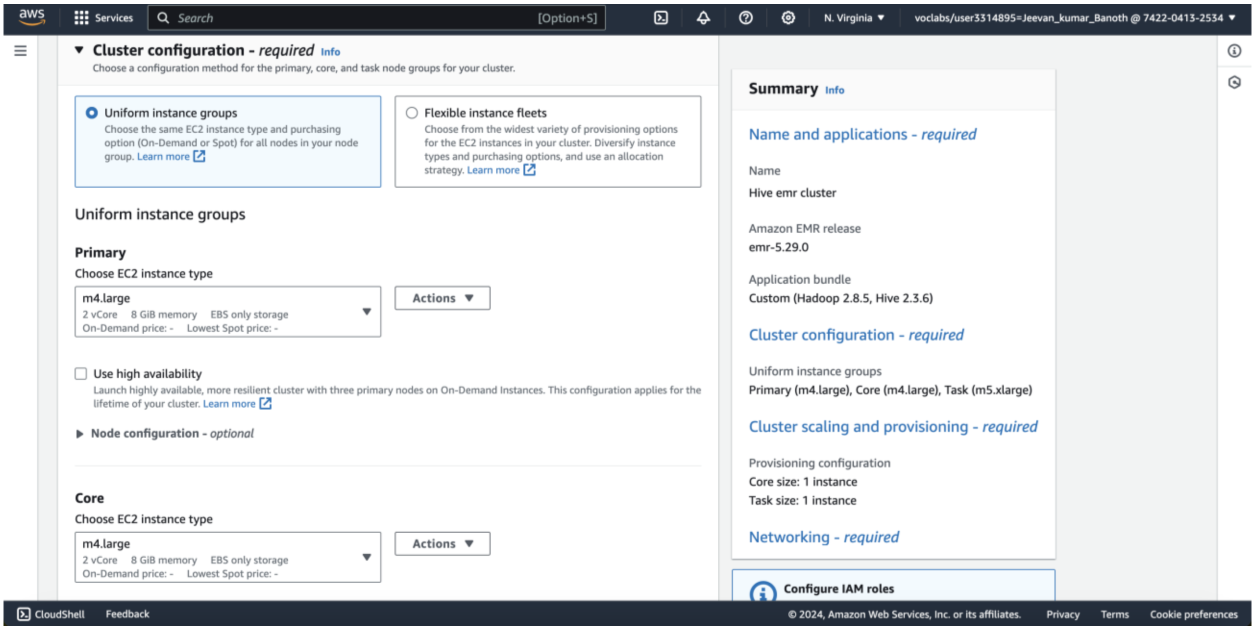
In Software Configuration, choose the Release option emr-5.29.0.

Verify that the following programs are selected:

Hadoop 2.8.5 2.3.6 Hive

Delete Hue and Pig as well as any other selected apps.

Leaving everything as it is, click Next.



Under the Cluster Nodes and Instances section, click the pencil icon next to the instance type for the primary node type, choose m4.large, and save.

To change the core node type instance type, click the pencil icon next to it, choose m4.large, and save.

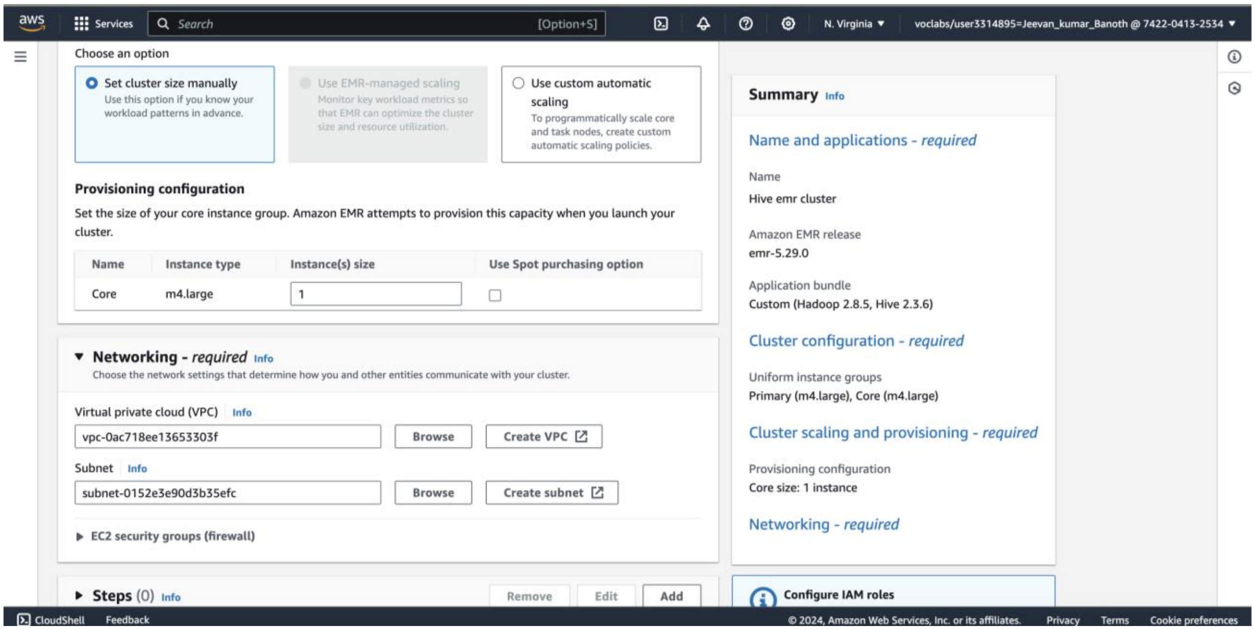
Verify that the following numbers of instances are set:

principal node: 1

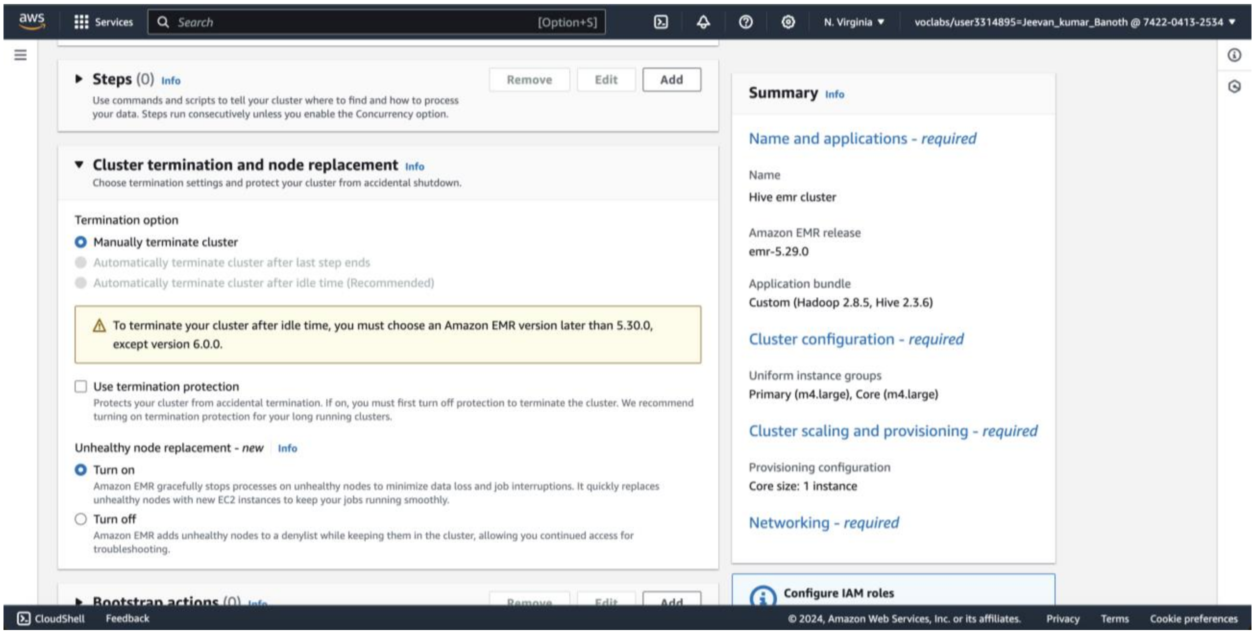
Two at the center node

Task node: 0

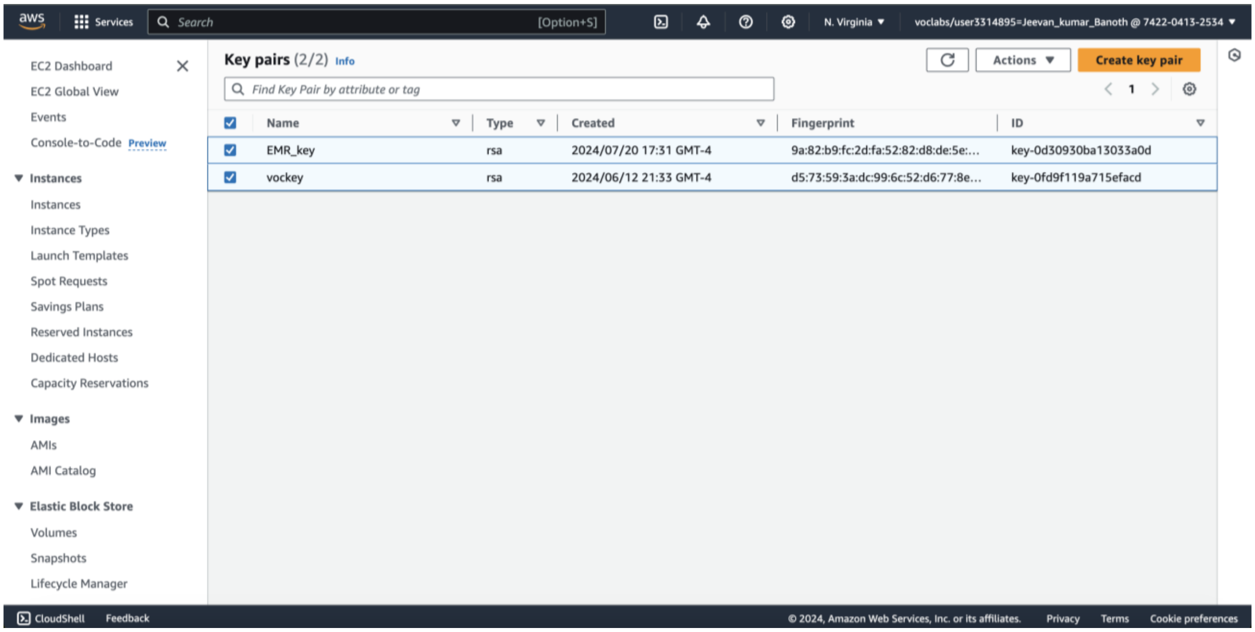
Leaving everything as it is, click Next.



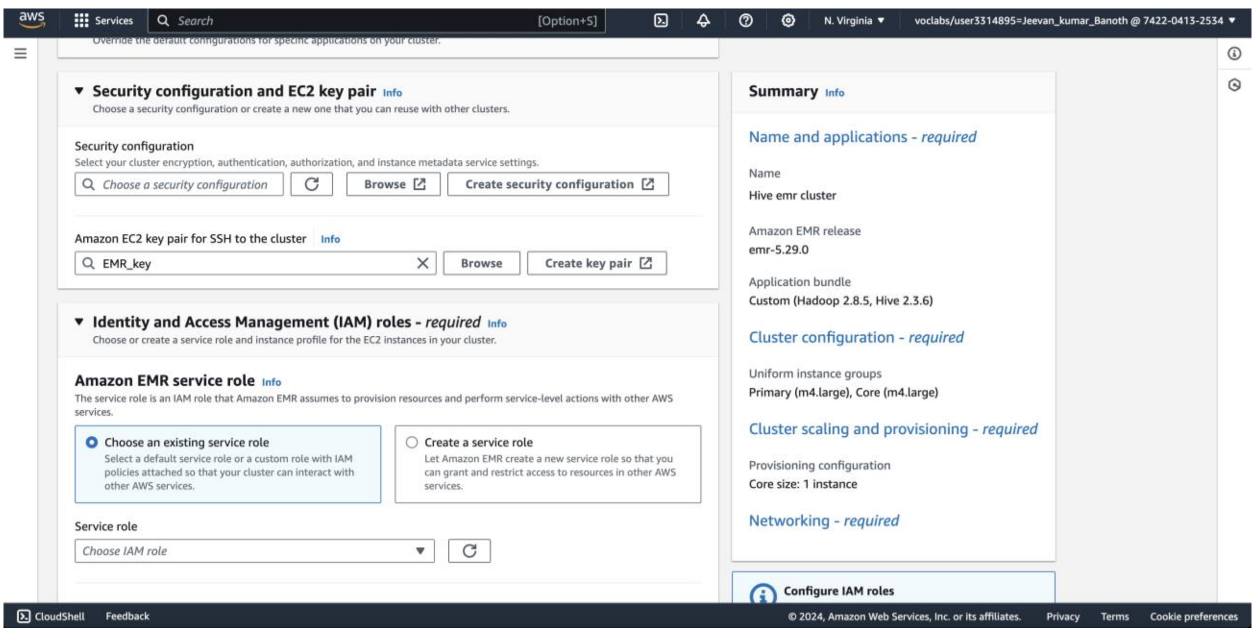
In choose options, set cluster size manually and in the network security as shown above for VPC and subnet.



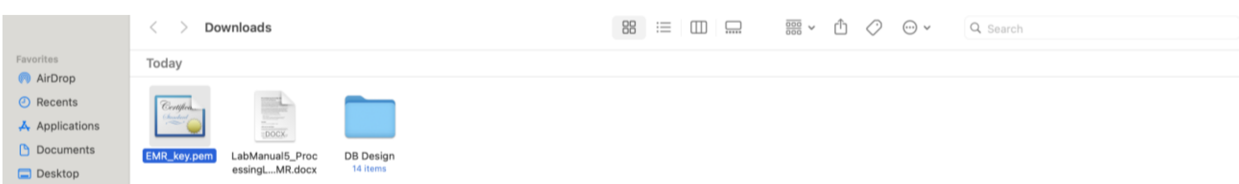
By expanding the cluster termination and node replacement section, for Termination option, select manually terminate cluster and Unhealthy node replacement turn it on.



Security Configuration: Pick the Vockey key pair for EC2 key pair. Select Custom for Permissions. Make sure that the EMR_DefaultRole is chosen for the EMR role.

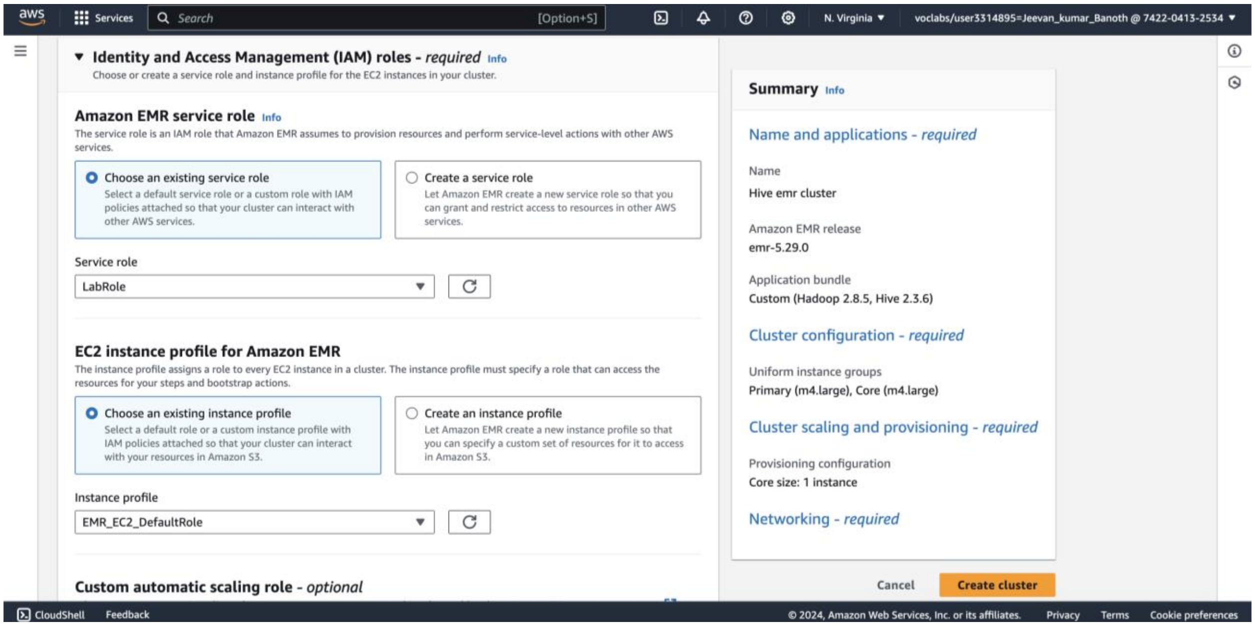


Make sure that the EMR_EC2_DefaultRole is chosen for the EC2 instance profile. For Auto Scaling Role, choose to continue without any role. Expand the EC2 Security groups section and use default settings.

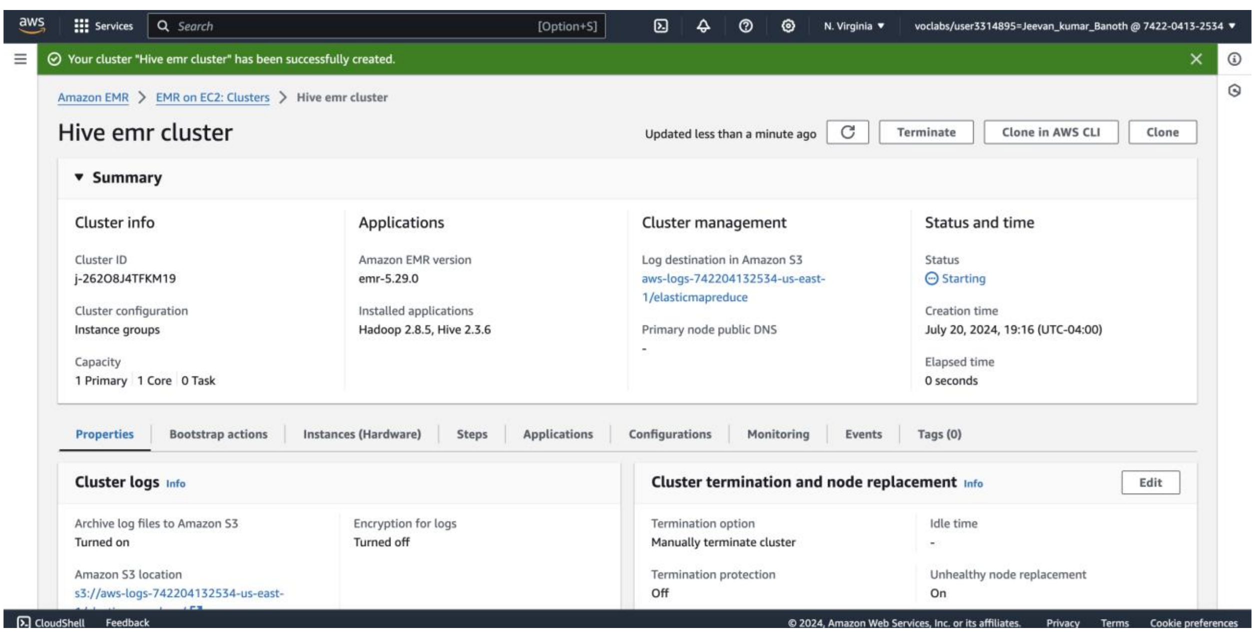


As soon as we choose the EMR_key, a .pem file will be downloaded in our local download's files as shown above.

We will use that for later in this lab.

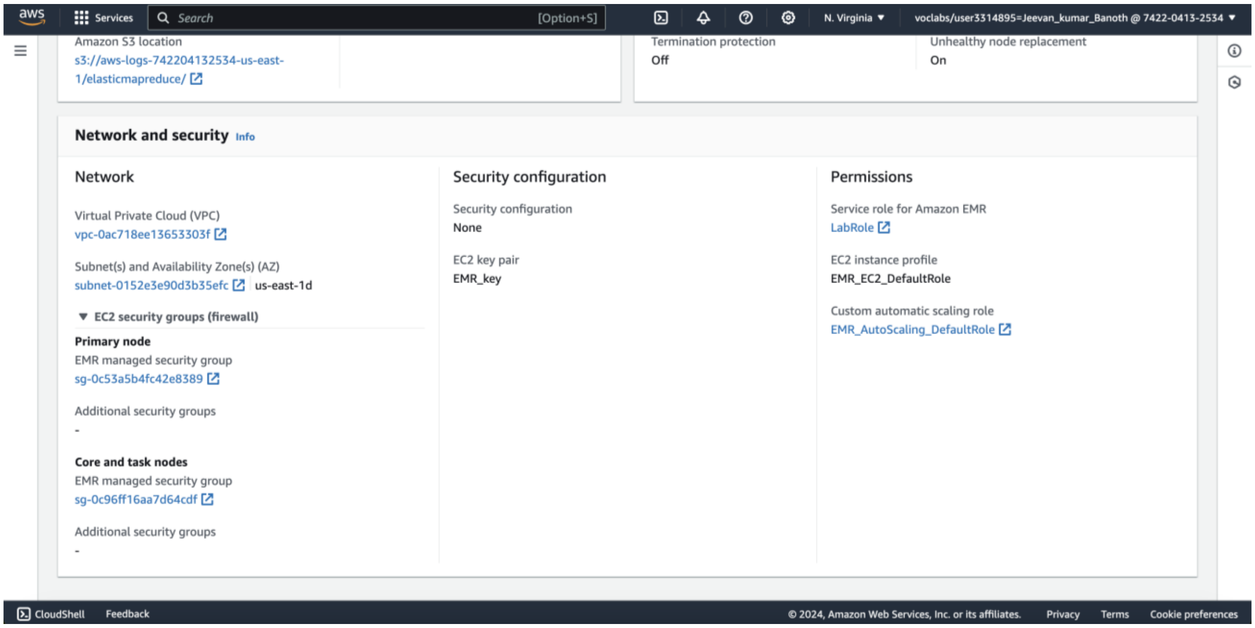


For Identity and Access management roles, we will be choosing the configuration as above.

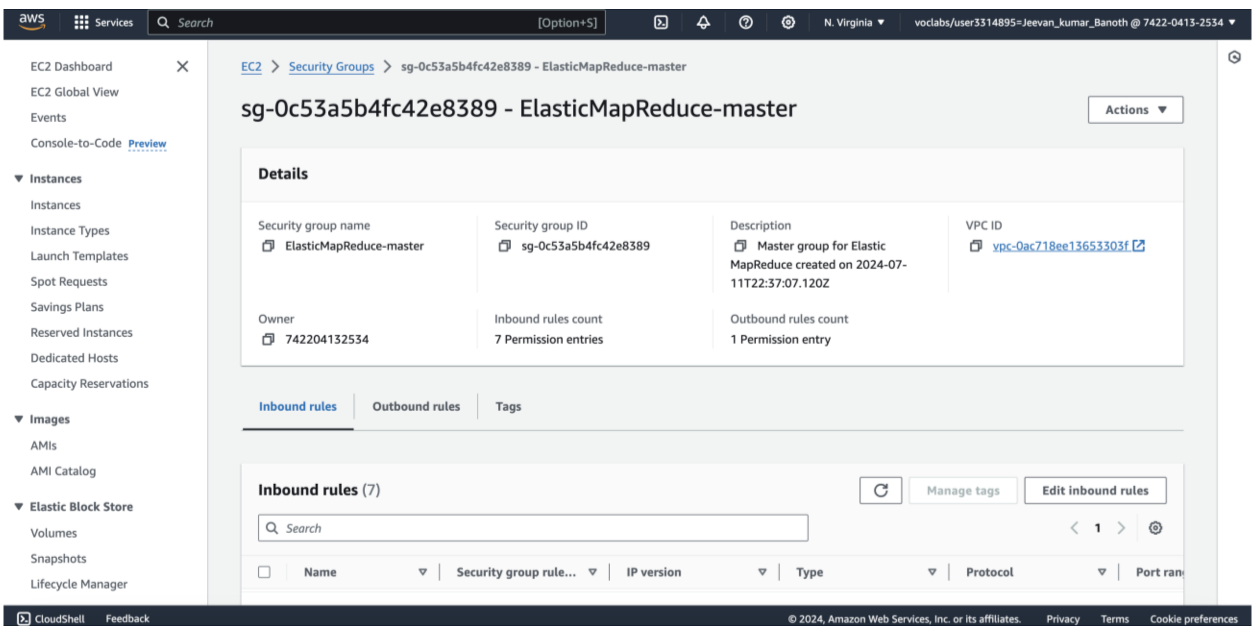


Then press create cluster, it will be in the starting status as shown above.

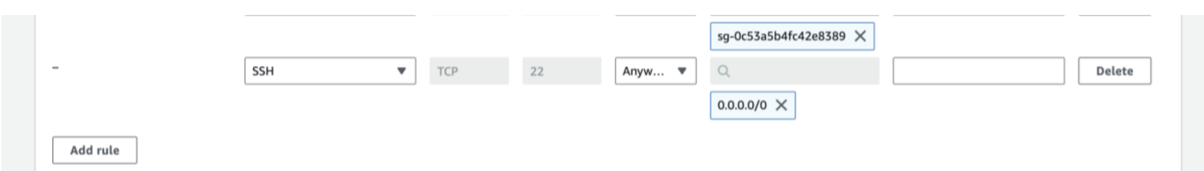
In the Network and security tab, we can see a primary node and there is a link under it, usually it starts with sg-0 something like this, as shown below.



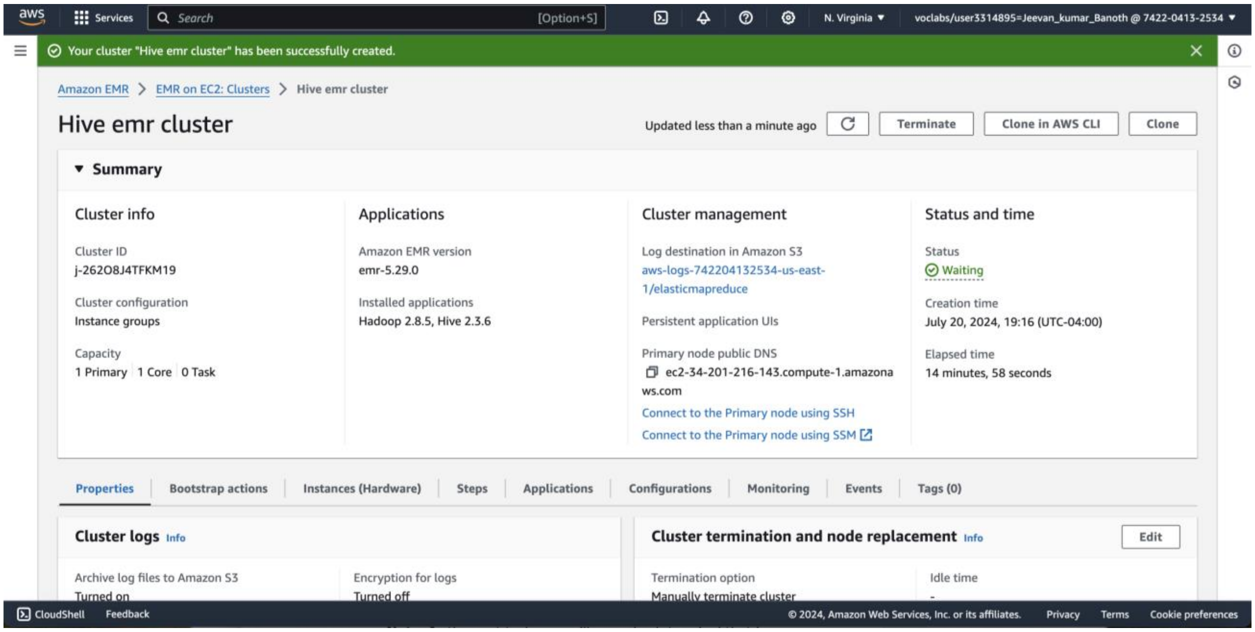
Click on that link, it will open in a different tab, as shown below.



Here, choose in the inbound rules as edit inbound rules and add rule in there as below.



In the place of type, choose SSH and followed by Anywhere IPv4 and save it. This gets added to our security group.

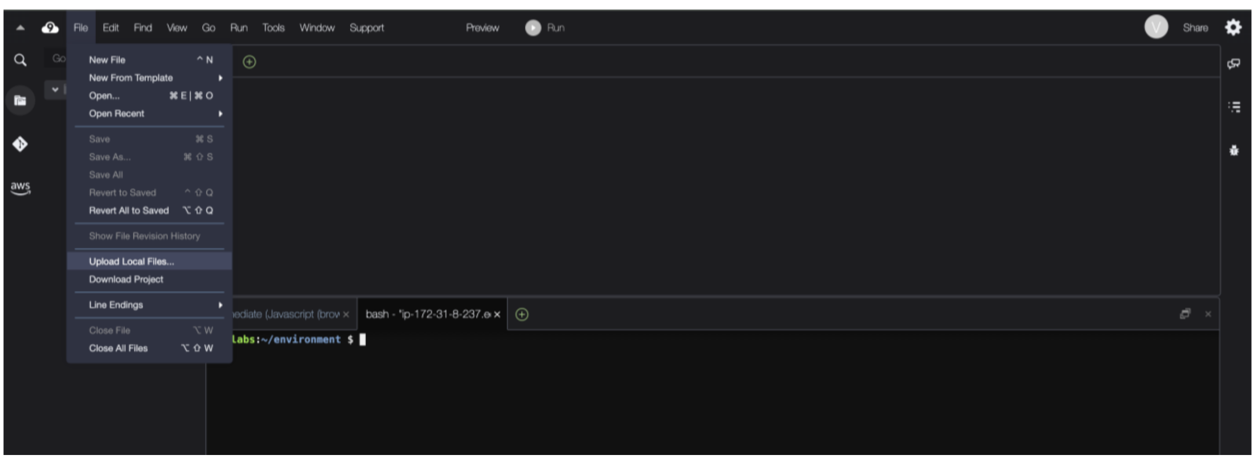


Now, as you can see above status get changes to waiting.

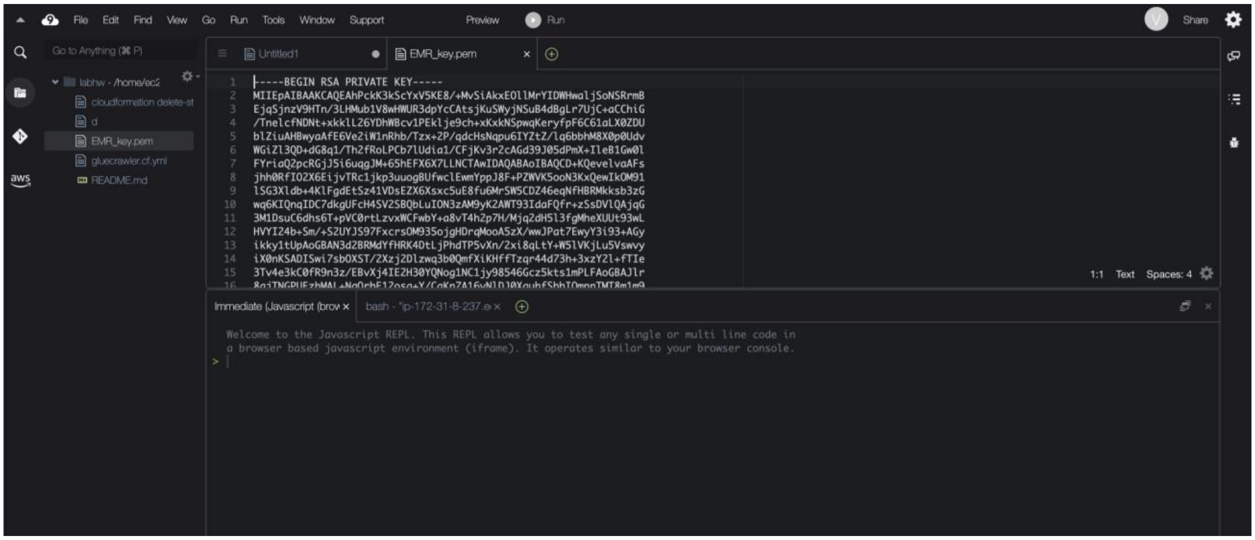
Task 2: Remote Connection to Hadoop Master via SSH

Locate the public DNS address assigned to the main node in the Cluster Summary tab and make a copy.

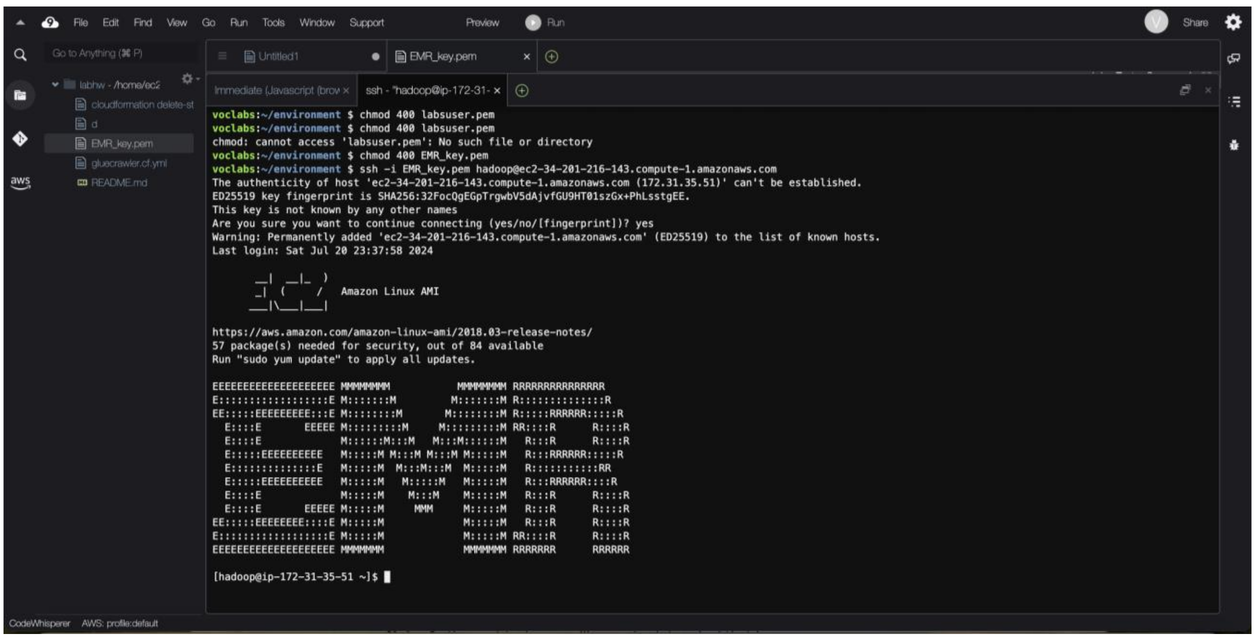
Next, go to your AWS Management Console and search for Cloud9 to open it. You will now select the Cloud9 Instance environment, then click on Open IDE. Now you create a new file from Cloud9 IDE by choosing File > New File. Lastly, paste this public DNS address in a document so you can use it later.



As you can see above, there is a upload butting, by using that we will going to upload the EMR_Key.pem file which we have downloaded earlier.



As you can see the file has been uploaded to cloud9 successfully.

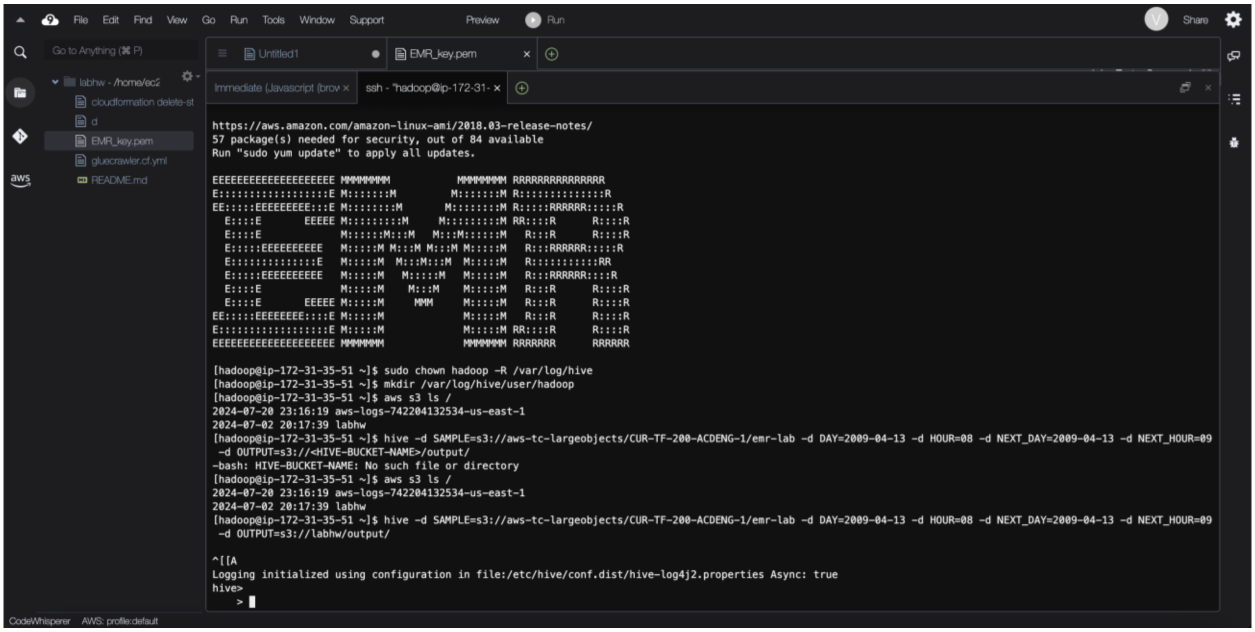


Now we will start examining the file by using the commands given in the lab manual.

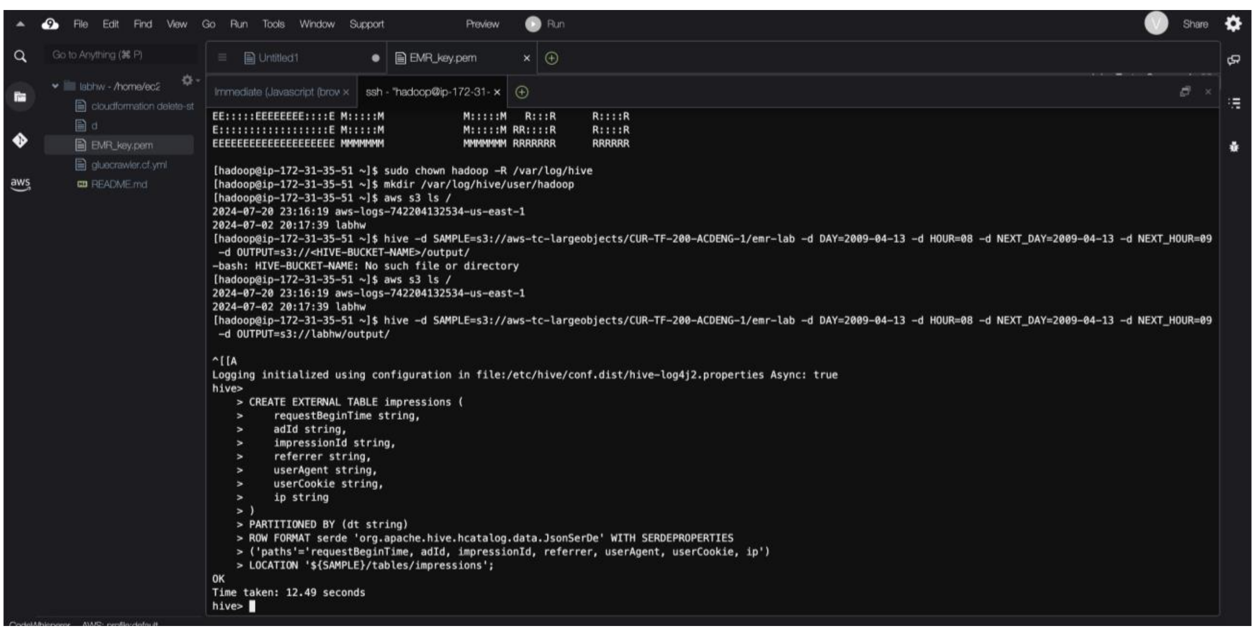
First, we will be changing the directory to EMR_key.pem file because that’s the file we are examining so.

Next, we will be computing it by using the ssh commands given by replacing the file name with what we gave.

In my case it is EMR_key.pem so We must change this according to the code.

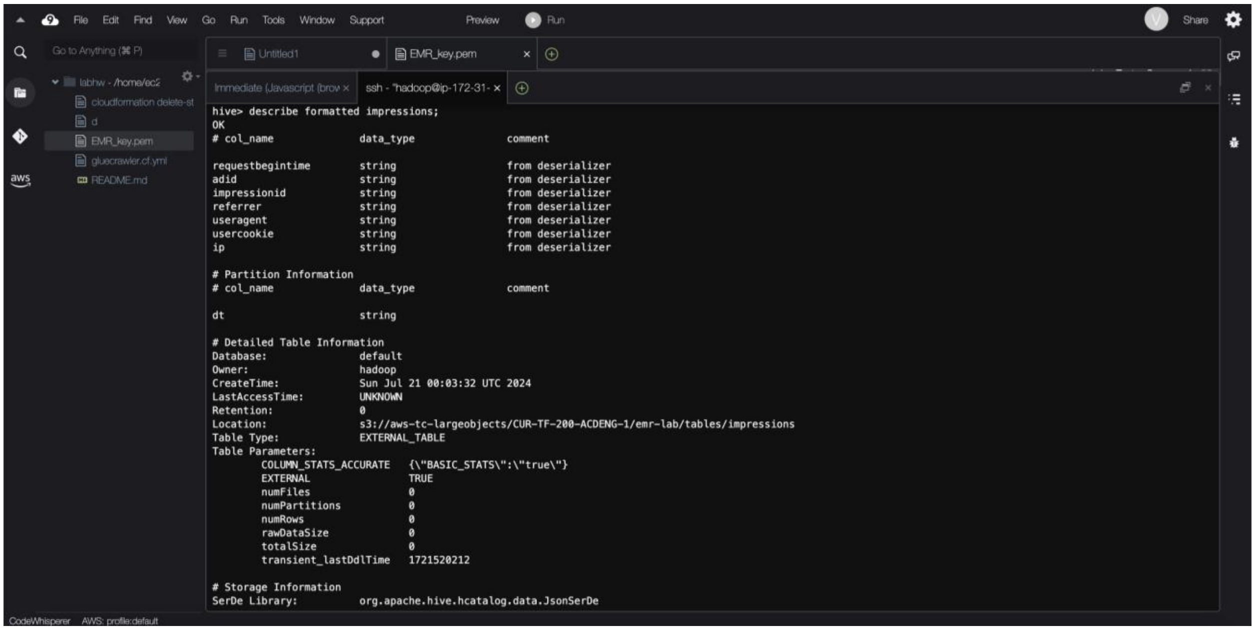


By using aws and hive commands we will find the bucket name that we are working on and replace that with the next code.

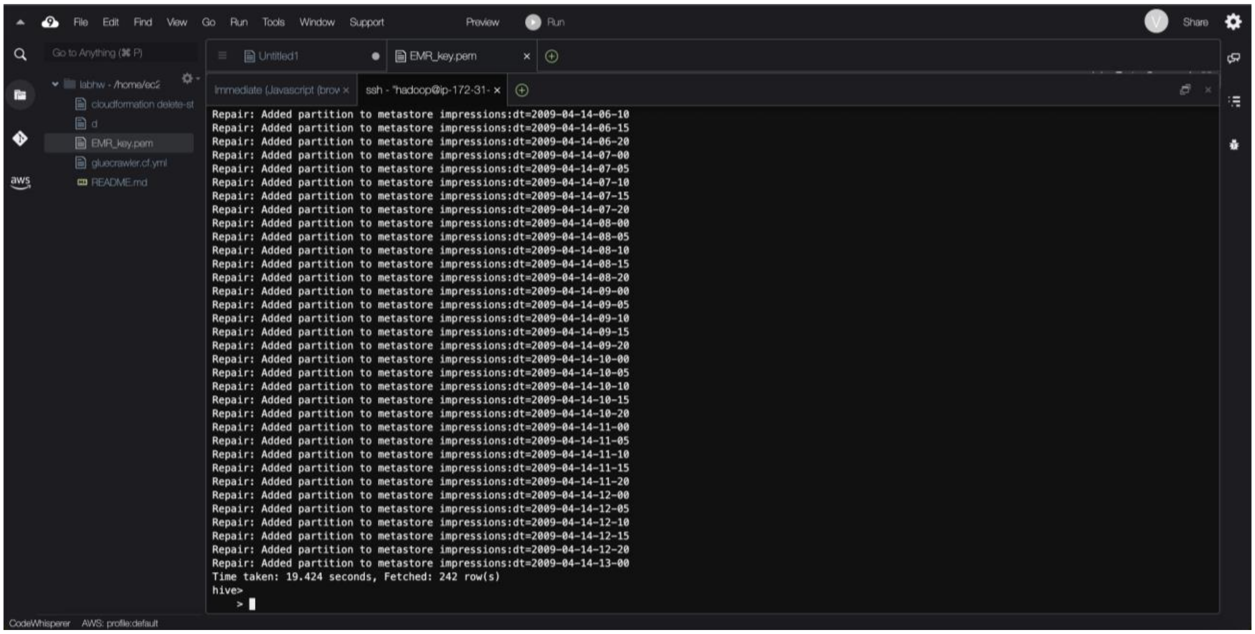


So, from that hive –d command, we will be going to be redirected into the hive terminal and we will work in that only from now on.

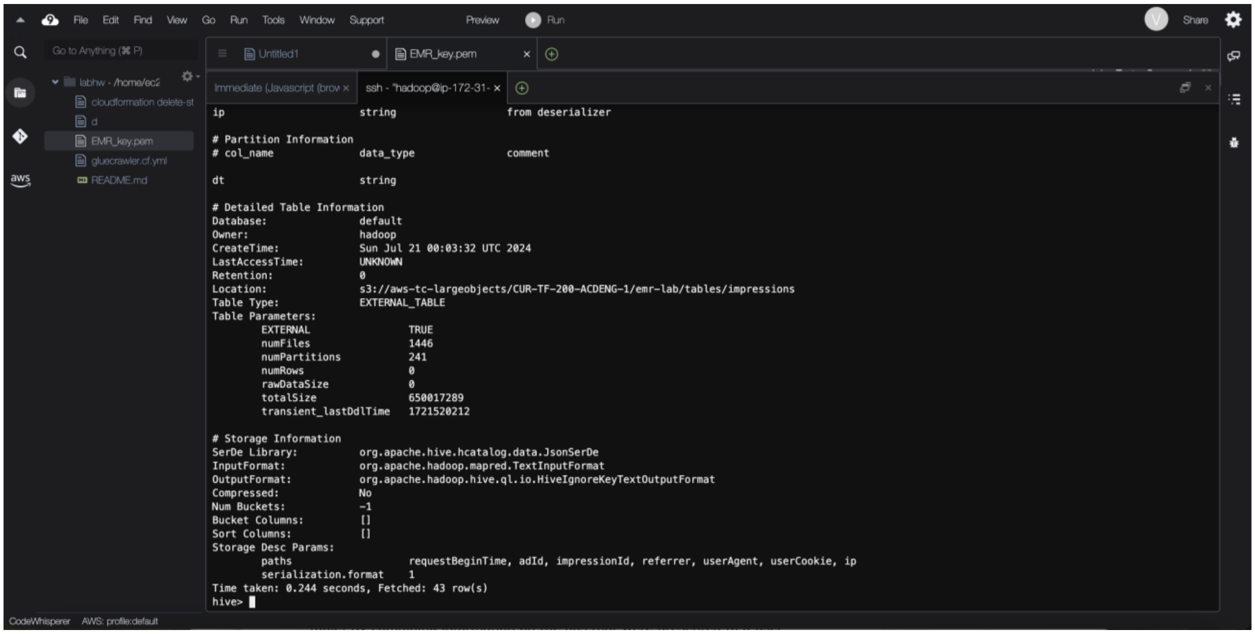
Next code is for creating the impressions table and time taken to run this is 12.49 seconds.



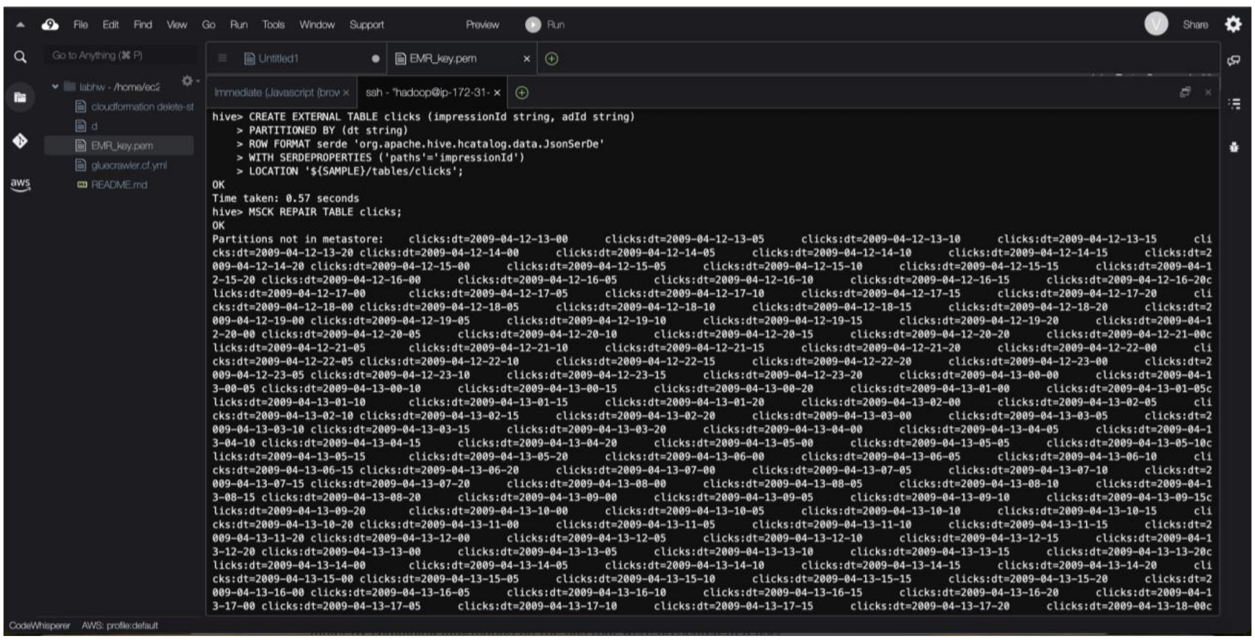
Next we will be updating the hive metadata for impressions by using the commands. Like for checking current partitions, repair table to add partitions and verifying the partitions.



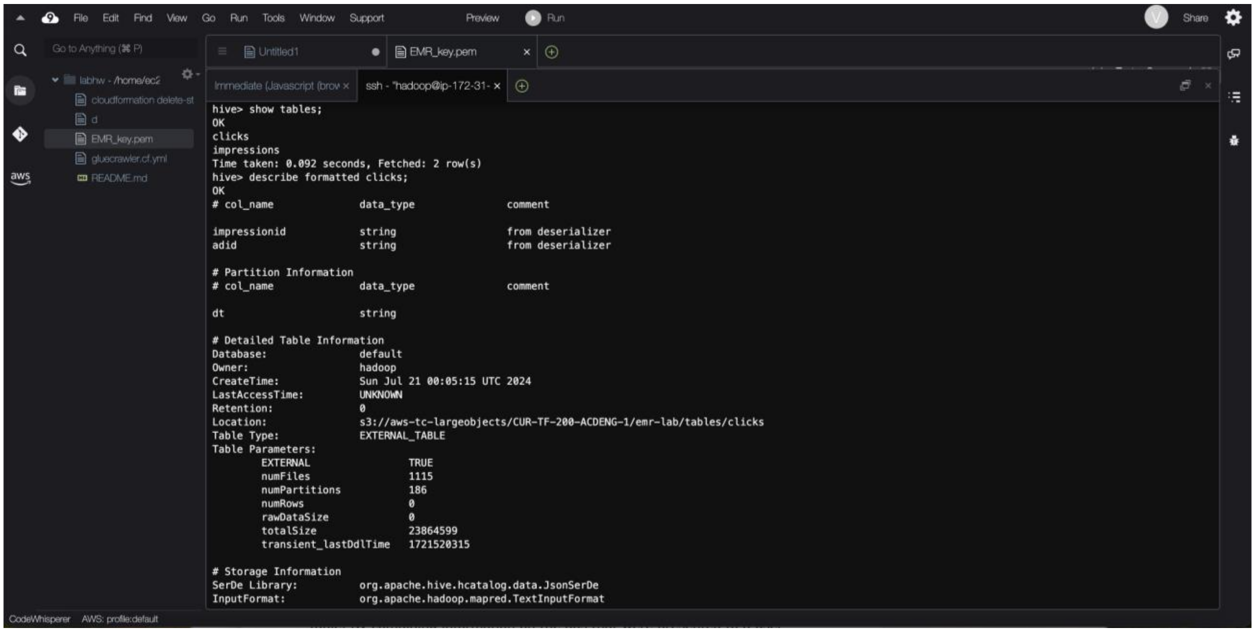
Next, we will be creating an external table for clicks.



Now, in this we will be updating the hive metadata for clicks and repairing the table to add partitions.

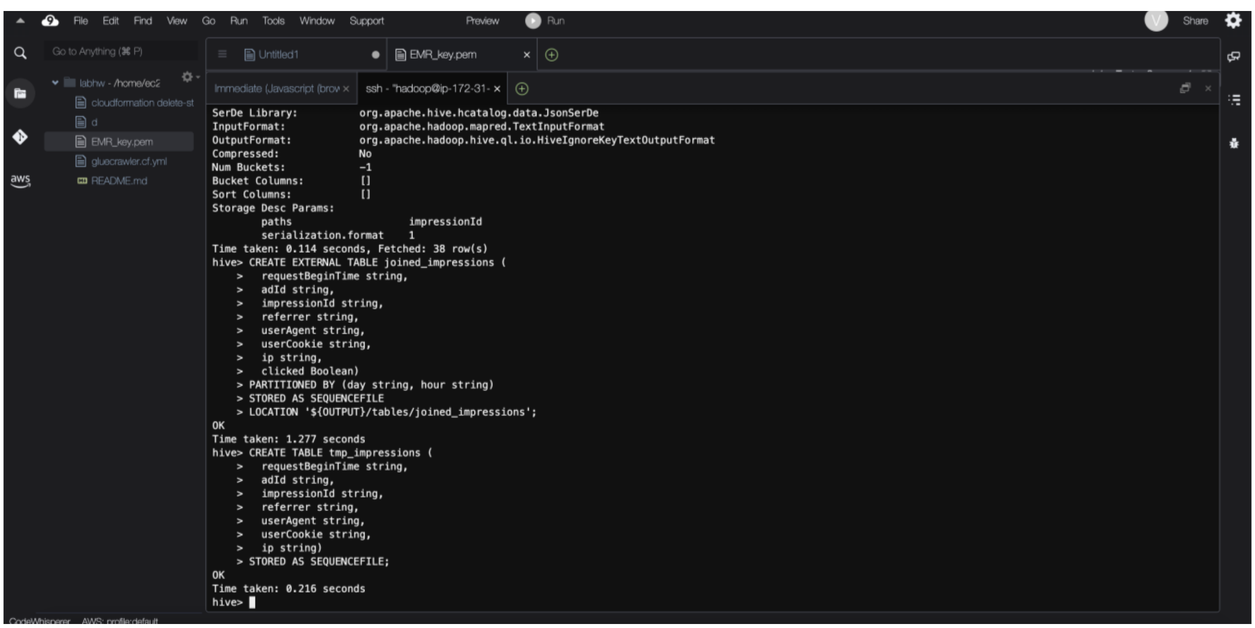


Then we will have to verify the tables. Like listing all the tables and describing the click in table.



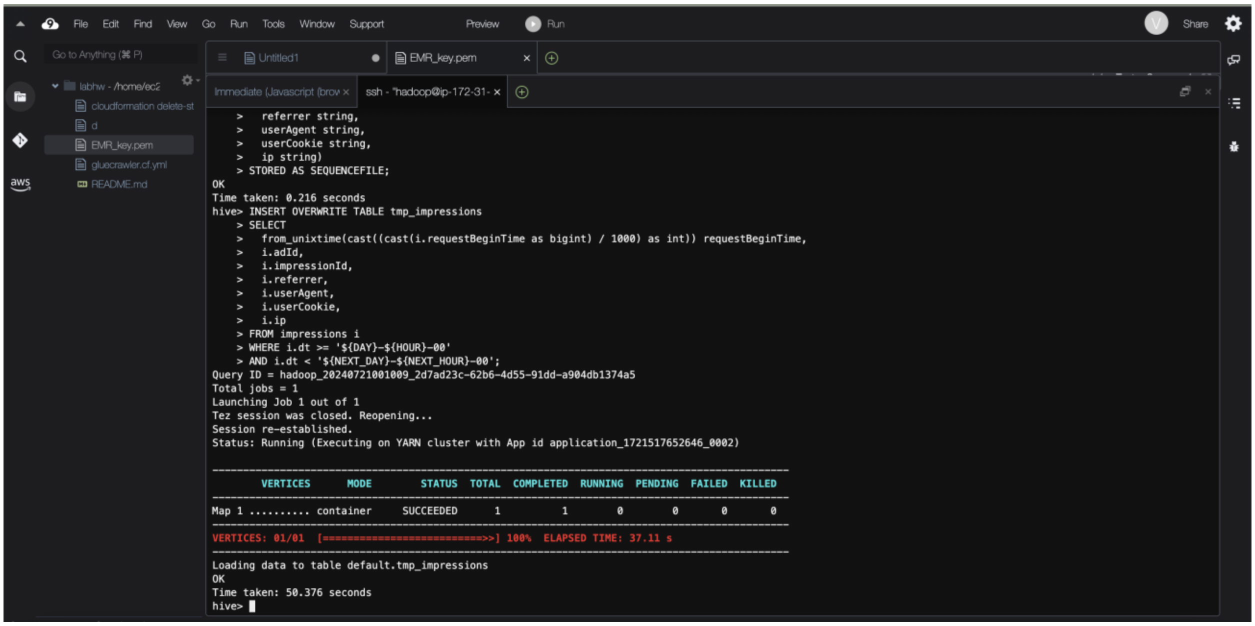
Task 5: Using Hive to Join Tables

In this work, you will create a new table called joined impressions by joining the impressions and clicks tables. This gives you insights into user behavior by combining data about impression information about advertising that users saw with clicks information about ads that users clicked on.

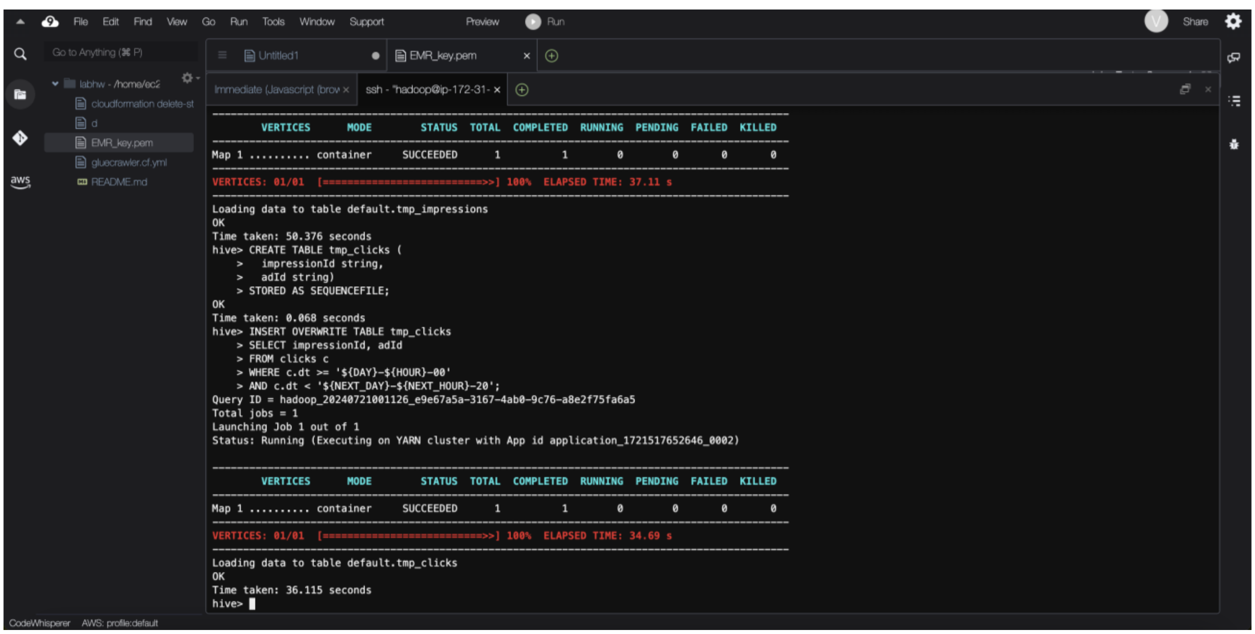


Now, we will be creating a joined impressions table. This table will have the same columns as the impressions and one more column which is clicked to indicate if the ad is clicked or not.

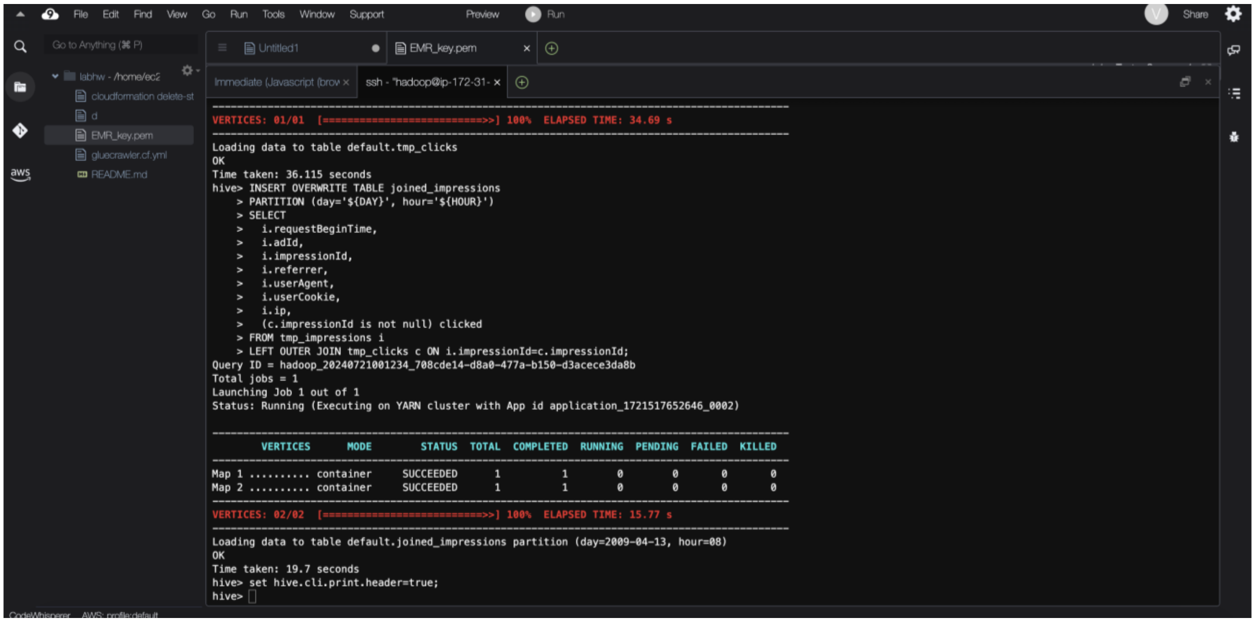
Creating a table for temporary impressions to store the data.



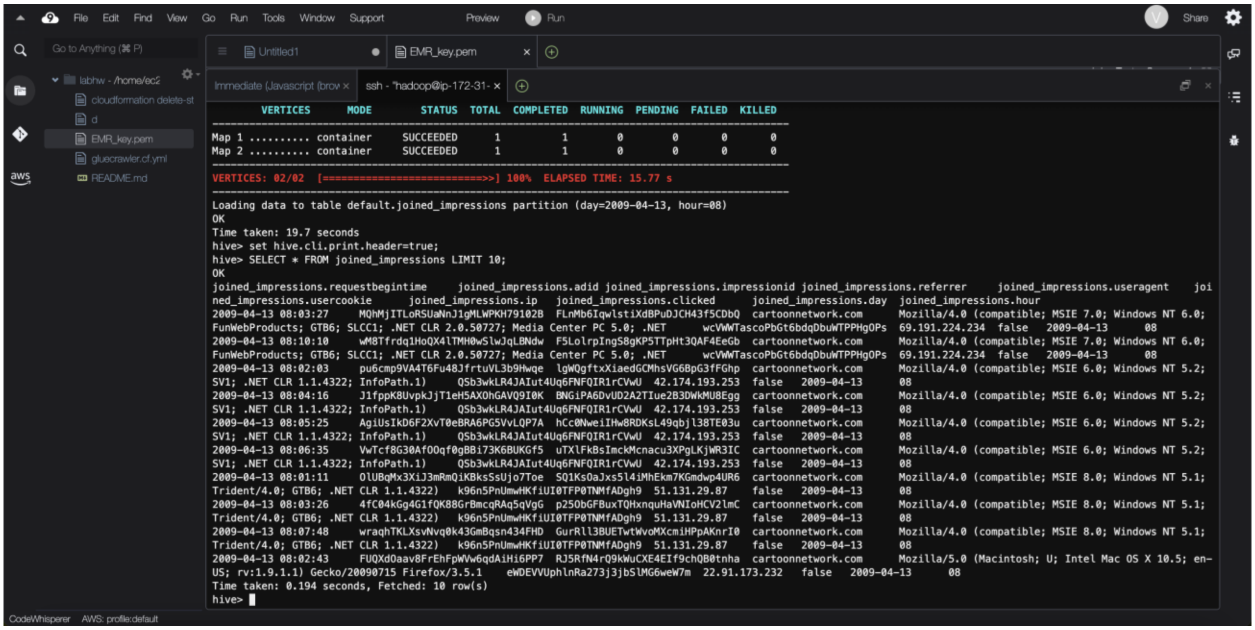
Inserting data into temp_impressions table for certain time duration. And the time took for this is 50.37 seconds.



Now, creating one more table for temporary clicks for storing the click data.

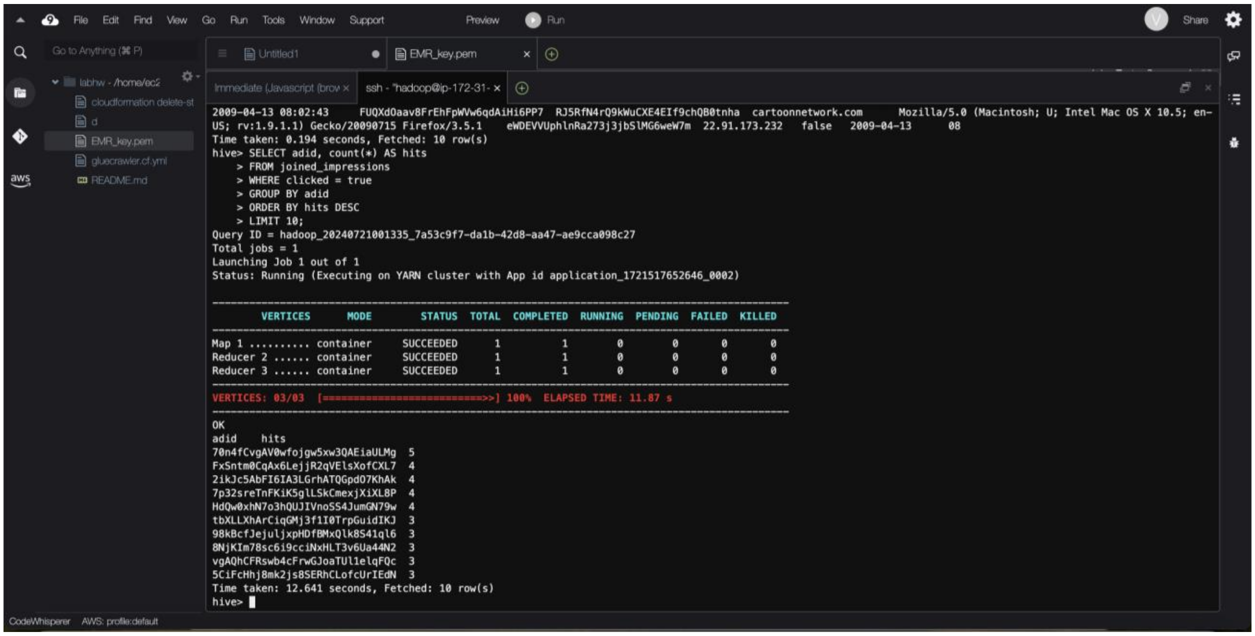


Now inserting the data into the temp_clicks table for slightly longer duration to see its performance.

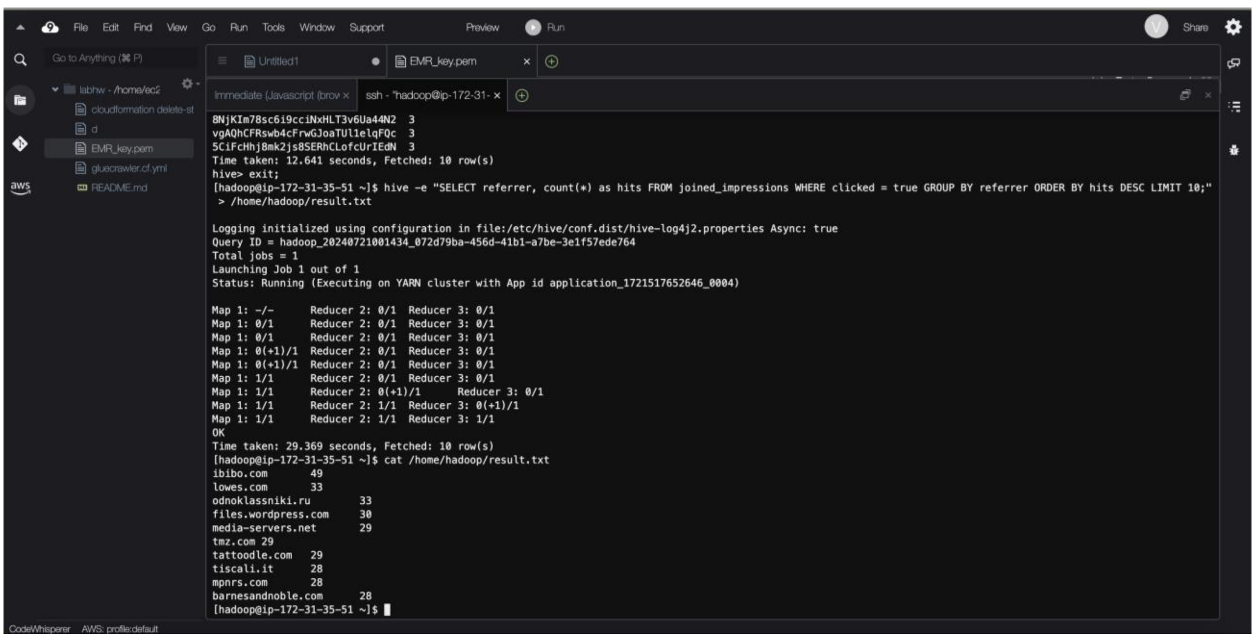


Performing the join of temp_impressions and temp_clicks table and writing the results into joined impressions table.

Using the hive commands, printing the column names as header.



To find the 10 most clicked ads IDs and to get the first 10 rows of from the joined_impressions table we will be using the above commands.



Now we will have to exit from the hive and save all that data into s3 bucket. We must find out the path and using cat commands we will be saving those results into that bucket at specified locations.

Conclusion:

In this laboratory we implemented the entire process of data warehousing with AWS EMR-HIVE machinery from padding of bare necessities needed to handle the infrastructure and finally using SQL-like queries to analyze the data.

At first, launching an EMR cluster was required by the laboratory which is important for processing large scale data. Through AWS Cloud9, you SSHed into the EMR cluster and configured a strong environment that supported all future activities.

These tables, which were stored in json format, were very relevant when it came to understanding user interaction with ads. Thus designing these two tables and telling where they will be located on S3's gives you foundation of good places for storage retrieval.

To elevate the performance of the datasets even higher, you turned these datasets into external tables with SequenceFile format. This transformation was made using HiveQL commands that took advantage of Hadoop's built-in compression features leading to great improvements in processing speediness.

The above move saw the creation of tables namely impressions_seq and clicks_seq which enables smooth data accessing thereby lowering space requirements. In keeping with the basic structure, your focus shifted to another essential duty of merging impressions data with clicks of information.

The aim was to combine such sources into a unified region where one could understand whose advertisements were clicked by users. To achieve this end goal, you set up a fresh external table known as joined_impressions intended to store all collected figures derived from both sources. An extra column was added to this table showing if an impression led to a click or not.

After you filled in the table called "joined impressions," you used Hive to query the data. While configuring Hive CLI to include column names in its output display, you wrote some queries that extracted and examined the data. The earliest return offered an initial short summary while the last ones revealed which ads were clicked on most frequently as well as from which sites or sources, this shed light on ad performance and how much users interacted with them.

You have shown during this lab that it is possible to handle big volume datasets, optimize data storage and perform complicated transformations and analysis using Hive or any other application running on EMR of Amazon.

By making sure that you put the resulting data into S3, you have ensured that they would always be there for future reference even when an EMR cluster is no longer running. Such an approach thus meets the expensive processing needs regarding data and cheap storage alternatives.